Incompressible Stokes Flow Annulus Benchmark Using Underworld3

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Abstract. We reproduce the annulus Stokes benchmark of Thieulot and Puckett (2018) (TP2018) using Underworld3 (UW3). The benchmark describes incompressible isoviscous Stokes flow in a cylindrical annulus with tangential velocity boundary conditions imposed on both the inner and outer boundaries, driven by radially directed gravity and analytically prescribed buoyancy forcing that generates convection cells parameterised by the harmonic number k. We investigate the accuracy and convergence behaviour of several mixed finite-element discretisations, including $P_1 \times P_0$, $P_1 \times P_1$, $P_2 \times P_1$, and $P_3 \times P_2$. Numerical solutions are evaluated against the analytical velocity and pressure fields using relative L_2-norm errors. The higher-order Taylor–Hood discretisations reproduce the expected convergence hierarchy for smooth Stokes solutions, with approximately third-order velocity and second-order pressure convergence for $P_2 \times P_1$, and fourth-order velocity and third-order pressure convergence for $P_3 \times P_2$. In contrast, lower-order discretisations exhibit reduced robustness, particularly in pressure recovery. These results demonstrate that UW3 accurately reproduces analytical Stokes solutions in curved geometries and provides a robust framework for future annulus geodynamics simulations.

1 Introduction

Accurate numerical solution of the incompressible Stokes equations is fundamental to modelling mantle convection, lithospheric deformation, subduction dynamics, and planetary evolution. Many geodynamic applications require simulations in curved geometries, such as annulus, spherical cap, and full spherical shell. These geometries introduce additional numerical challenges associated with curved boundaries, pressure nullspaces, coordinate-dependent forcing terms, and tangential boundary conditions.

Reproducing analytical benchmarks remains one of the most rigorous approaches for assessing the correctness of numerical implementations. Unlike code-to-code comparisons, analytical benchmarks enable direct quantification of discretisation errors and convergence rates. They are particularly valuable for geodynamic Stokes problems because they simultaneously assess velocity–pressure coupling, incompressibility preservation, boundary condition implementation, and the accurate representation of curved geometries (Kramer et al., 2021; Davies et al., 2022).

The annulus benchmark introduced by TP2018 provides an exact analytical solution for incompressible isoviscous Stokes flow in a cylindrical shell subjected to radially directed gravity. The benchmark was specifically designed for geodynamics applications and has since been widely used for verification of geodynamic codes within the community (Gassmöller et al., 2024).

Unlike classical Cartesian manufactured solutions, the benchmark incorporates curved inner and outer boundaries, non-trivial pressure distributions along the boundaries, coordinate-dependent buoyancy forcing, tangential velocity boundary conditions, and controllable azimuthal complexity through the harmonic parameter k. These characteristics make the benchmark particularly well suited for assessing finite-element formulations in annulus and curved-domain mantle convection problems.

UW3 is a next-generation geodynamics framework built around PETSc and symbolic problem specification in Python. In this study, we reproduce the TP2018 annulus benchmark to verify the curved-domain finite-element implementations in UW3. We evaluate the convergence and accuracy of several mixed finite-element discretisations, including both lower-order and higher-order velocity–pressure element pairs. Numerical results are compared against the analytical velocity and pressure fields using L_2-norm errors and boundary pressure diagnostics. Particular attention is given to the convergence behaviour of Taylor–Hood discretisations and the performance of lower-order and equal-order element formulations.

2 Governing Equations

We consider steady state incompressible Stokes equations:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

The stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta \boldsymbol{\varepsilon}(\mathbf{u}), \tag{2}$$

with viscosity η and strain-rate tensor

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right). \tag{3}$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration.

3 Analytical Benchmark Solution

Following Thieulot and Puckett (2018), the analytical velocity field in polar coordinates is defined as

$$v_\theta(r,\theta) = f(r)\cos(k\theta), \tag{4a}$$

$$v_r(r,\theta) = kg(r)\sin(k\theta). \tag{4b}$$

where

$$f(r) = Ar + \frac{B}{r}, \tag{5a}$$

$$g(r) = \frac{A}{2}r + \frac{B}{r}\ln r + \frac{C}{r}. \tag{5b}$$

The constants A and B are chosen such that radial velocity vanishes at both annulus boundaries:

$$A = -C\frac{2(\ln R_1 - \ln R_2)}{R_2^2 \ln R_1 - R_1^2 \ln R_2}, \tag{6a}$$

$$B = -C\frac{R_2^2 - R_1^2}{R_2^2 \ln R_1 - R_1^2 \ln R_2}. \tag{6b}$$

The pressure field is

$$p(r,\theta) = kh(r)\sin(k\theta) + \rho_0(R_2 - r), \tag{7}$$

where

$$h(r) = \frac{2g(r) - f(r)}{r}. \tag{8}$$

The corresponding density field is

$$\rho(r,\theta) = kM(r)\sin(k\theta) + \rho_0, \tag{9}$$

with

$$M(r) = g''(r) - \frac{g'(r)}{r} - \frac{g(r)}{r^2}(k^2 - 1) + \frac{f(r)}{r^2} + \frac{f'(r)}{r}. \tag{10}$$

The benchmark calculations use the standard parameter values

$$R_1 = 1, \qquad R_2 = 2, \qquad C = -1, \qquad \rho_0 = 0.$$

where R_1 is the inner radius and R_2 is the outer radius. Gravity acts radially inward:

$$\mathbf{g} = -\mathbf{e}_r. \tag{11}$$

The benchmark assumes constant viscosity $\eta = 1$ and prescribes analytical velocity on boundaries, and density field in the domain. Increasing the harmonic parameter k increases azimuthal complexity and produces $2k$ convection cells in the annulus. This enables systematic evaluation of numerical performance across increasingly oscillatory flow structures.

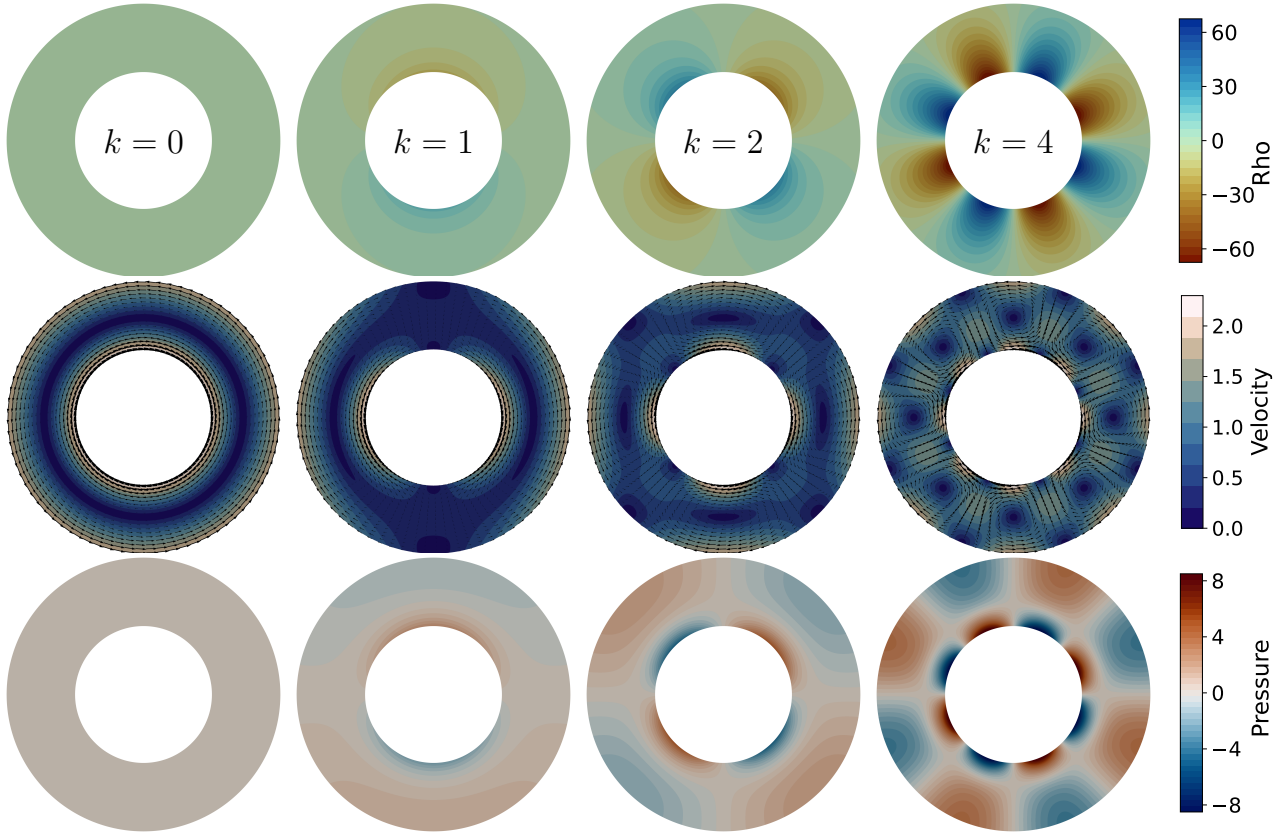

Figure 1. Analytical density, velocity, and pressure fields. Columns correspond to increasing k, which increases the number of convection cells in the annulus.

4 Numerical Methodology

4.1 UW3 Formulation

The benchmark is implemented in UW3 using PETSc-based finite-element Stokes system. The governing equations are assembled symbolically using SymPy and solved through PETSc Krylov subspace methods. A constant-pressure nullspace is supplied to PETSc to remove the pressure gauge ambiguity. Following solution, the domain-average numerical pressure is removed prior to error evaluation. The annulus meshes are generated using unstructured triangular elements with progressively refined nominal cell sizes. The mesh geometry degree is set to linear for all reported calculations.

4.2 Finite-Element Discretisations

The following mixed finite-element pairs are evaluated:

Velocity space	Pressure space	Description
P_1	P_0	Linear velocity, discontinuous constant pressure
P_1	P_1	Equal-order continuous discretisation
P_2	P_1	Taylor–Hood discretisation
P_3	P_2	Higher-order Taylor–Hood discretisation

The quadrature degree is chosen according to the highest polynomial order present in the discretisation. For smooth incompressible Stokes solutions, stable Taylor–Hood-type discretisations are expected to exhibit velocity convergence of order $p+1$ and pressure convergence of order p for a $P_{p+1} \times P_p$ discretisation (Girault and Raviart, 1986; Boffi et al., 2013).

4.3 Error Metrics

Relative L_2-norm errors are computed as

$$\|e_q\|_{L_2,\mathrm{rel}} = \left(\frac{\int_\Omega |q_h - q|^2 \, d\Omega}{\int_\Omega |q|^2 \, d\Omega} \right)^{1/2}, \tag{12}$$

where q_h is the numerical solution and q is the analytical solution. Convergence rates are computed from successive mesh refinements.

add convergence rate formula. add absolute l2 norms.

5 Results

5.1 Convergence Behaviour

The convergence behaviour for all tested discretisations is shown in Fig. 2. The $P_2 \times P_1$ Taylor–Hood discretisation exhibits approximately third-order velocity convergence and second-order pressure convergence, consistent with theoretical expectations for smooth Stokes solutions.

The higher-order $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity convergence and third-order pressure convergence on coarse-to-intermediate mesh resolutions. At the finest resolutions, convergence rates begin to plateau, likely reflecting accumulated effects from geometry approximation, quadrature errors, and solver tolerances.

The lower-order discretisations show reduced robustness. The $P_1 \times P_0$ pair fails to produce reliable convergence behaviour for this benchmark, while the equal-order $P_1 \times P_1$ discretisation produces acceptable velocity convergence but weaker and less consistent pressure convergence. These observations are consistent with established finite-element theory for incompressible Stokes flow (Boffi et al., 2013; Thieulot and Bangerth, 2022).

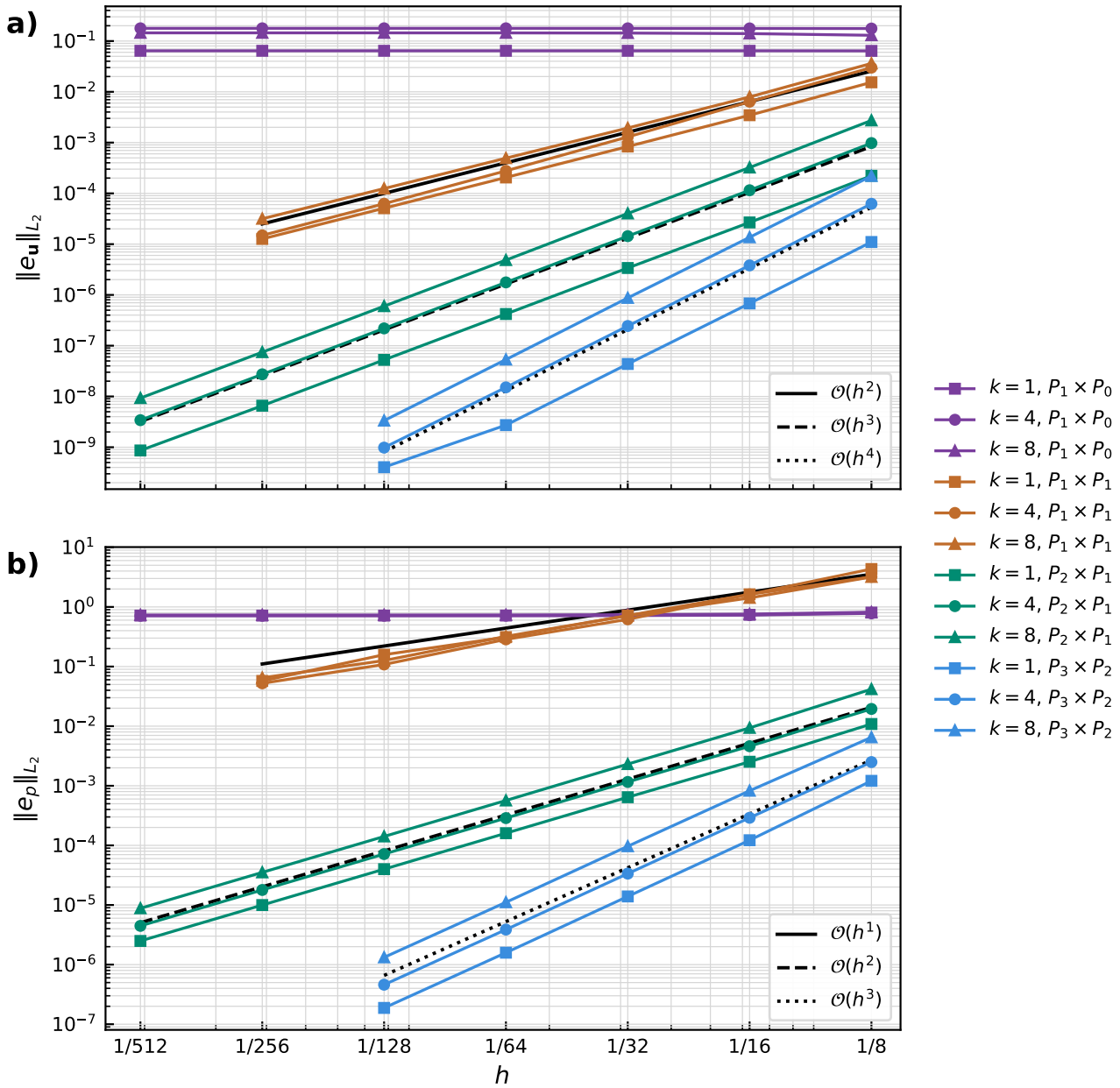

Figure 2. Convergence of relative volume L_2-norm velocity and pressure errors for the annulus Thieulot benchmark. Reference lines indicate the expected smooth-solution rates for the higher-order element pairs.

Table 1. Errors and Convergence Rates for $P_1 \times P_0$.

	$k=1$				$k=4$				$k=8$			
h	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate
$1/8$	6.398e-02	–	8.040e-01	–	1.763e-01	–	7.800e-01	–	1.298e-01	–	8.163e-01	–
$1/16$	6.427e-02	-0.01	7.384e-01	0.12	1.775e-01	-0.01	7.295e-01	0.10	1.401e-01	-0.11	7.589e-01	0.11
$1/32$	6.426e-02	0.00	7.220e-01	0.03	1.780e-01	-0.00	7.161e-01	0.03	1.435e-01	-0.04	7.403e-01	0.04
$1/64$	6.426e-02	0.00	7.177e-01	0.01	1.781e-01	-0.00	7.124e-01	0.01	1.445e-01	-0.01	7.349e-01	0.01
$1/128$	6.425e-02	0.00	7.167e-01	0.00	1.782e-01	-0.00	7.114e-01	0.00	1.447e-01	-0.00	7.334e-01	0.00
$1/256$	6.425e-02	0.00	7.164e-01	0.00	1.782e-01	-0.00	7.112e-01	0.00	1.448e-01	-0.00	7.330e-01	0.00
$1/512$	6.425e-02	0.00	7.163e-01	0.00	1.782e-01	-0.00	7.111e-01	0.00	1.448e-01	-0.00	7.329e-01	0.00

Table 2. Errors and Convergence Rates for $P_1 \times P_1$.

	$k=1$				$k=4$				$k=8$			
h	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate
$1/8$	1.537e-02	–	4.325e+00	–	2.994e-02	–	3.202e+00	–	3.629e-02	–	3.184e+00	–
$1/16$	3.459e-03	2.15	1.628e+00	1.41	6.385e-03	2.23	1.625e+00	0.98	7.925e-03	2.19	1.417e+00	1.17
$1/32$	8.358e-04	2.05	7.174e-01	1.18	1.293e-03	2.30	6.195e-01	1.39	1.937e-03	2.03	7.144e-01	0.99
$1/64$	2.056e-04	2.02	3.115e-01	1.20	2.776e-04	2.22	2.855e-01	1.12	4.931e-04	1.97	3.210e-01	1.15
$1/128$	5.095e-05	2.01	1.582e-01	0.98	6.252e-05	2.15	1.083e-01	1.40	1.243e-04	1.99	1.260e-01	1.35
$1/256$	1.266e-05	2.01	5.740e-02	1.46	1.487e-05	2.07	5.231e-02	1.05	3.158e-05	1.98	6.476e-02	0.96

Table 3. Errors and Convergence Rates for $P_2 \times P_1$.

	$k=1$				$k=4$				$k=8$			
h	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate
$1/8$	2.218e-04	–	1.087e-02	–	9.781e-04	–	1.944e-02	–	2.731e-03	–	4.174e-02	–
$1/16$	2.675e-05	3.05	2.530e-03	2.10	1.151e-04	3.09	4.611e-03	2.08	3.225e-04	3.08	9.354e-03	2.16
$1/32$	3.388e-06	2.98	6.427e-04	1.98	1.437e-05	3.00	1.159e-03	1.99	4.016e-05	3.01	2.310e-03	2.02
$1/64$	4.205e-07	3.01	1.601e-04	2.01	1.763e-06	3.03	2.869e-04	2.01	4.850e-06	3.05	5.666e-04	2.03
$1/128$	5.254e-08	3.00	3.992e-05	2.00	2.190e-07	3.01	7.152e-05	2.00	6.001e-07	3.01	1.412e-04	2.00
$1/256$	6.575e-09	3.00	9.979e-06	2.00	2.730e-08	3.00	1.787e-05	2.00	7.449e-08	3.01	3.524e-05	2.00
$1/512$	8.632e-10	2.93	2.494e-06	2.00	3.412e-09	3.00	4.466e-06	2.00	9.287e-09	3.00	8.802e-06	2.00

Table 4. Errors and Convergence Rates for $P_3 \times P_2$.

	$k=1$				$k=4$				$k=8$			
h	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate	$\lvert e_v\rvert_2$	rate	$\lvert e_p\rvert_2$	rate
$1/8$	1.102e-05	–	1.210e-03	–	6.233e-05	–	2.498e-03	–	2.198e-04	–	6.495e-03	–
$1/16$	6.792e-07	4.02	1.220e-04	3.31	3.811e-06	4.03	2.905e-04	3.10	1.363e-05	4.01	8.249e-04	2.98
$1/32$	4.360e-08	3.96	1.392e-05	3.13	2.441e-07	3.96	3.358e-05	3.11	8.723e-07	3.97	9.664e-05	3.09
$1/64$	2.729e-09	4.00	1.585e-06	3.13	1.510e-08	4.01	3.870e-06	3.12	5.323e-08	4.03	1.110e-05	3.12
$1/128$	4.043e-10	2.75	1.877e-07	3.08	9.826e-10	3.94	4.601e-07	3.07	3.344e-09	3.99	1.322e-06	3.07

5.2 Boundary Pressure Recovery

Pressure profiles extracted along the inner and outer annulus boundaries are compared against analytical solutions in Fig. 3. The higher-order Taylor–Hood discretisations reproduce the analytical pressure distributions accurately across the annulus boundaries, including extrema near regions of high azimuthal variation. Pressure discrepancies are most pronounced for lower-order discretisations and become increasingly visible for larger k, where sharper angular pressure gradients must be resolved.

Because the benchmark includes non-trivial pressure variation directly on curved boundaries, these comparisons provide a stringent test of both pressure interpolation quality and curved-domain numerical accuracy.

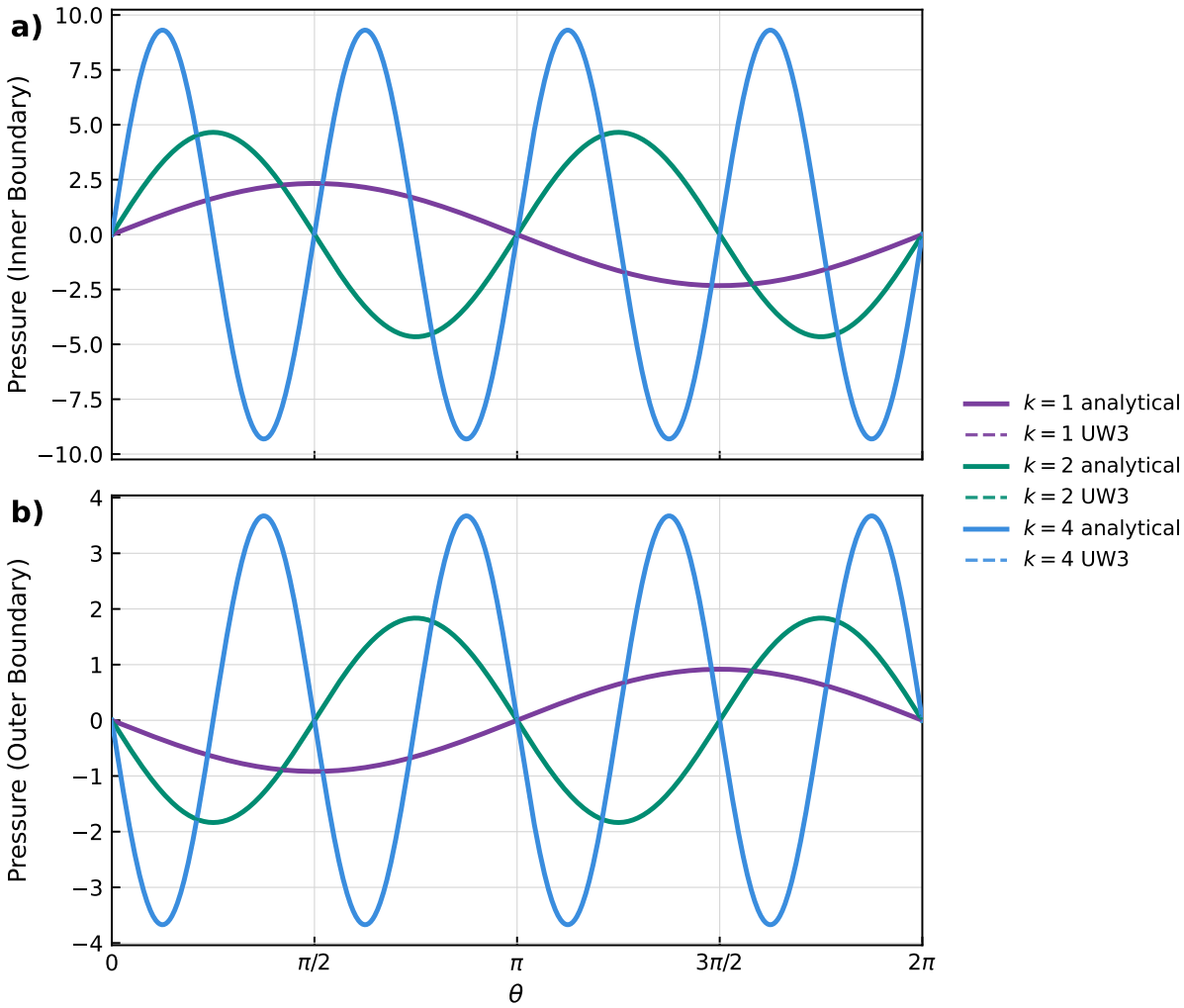

Figure 3. Pressure along the inner and outer annulus boundaries for $k = 1, 2, 4$, using the $P_2 \times P_1$, $h = 1/128$ calculations. Solid curves denote the analytical solution and dashed curves denote UW3 values sampled from the numerical pressure field.

5.3 Implications for Spherical Geodynamics

Although the benchmark is formulated in a two-dimensional annulus, its numerical characteristics are directly relevant to spherical-shell mantle convection calculations. In particular, the benchmark tests curved-boundary finite-element behaviour, radial body-force implementation, pressure nullspace treatment, incompressibility preservation, and higher-order mixed discretisations. These are all essential ingredients in regional and global spherical geodynamics models.

The strong convergence behaviour observed for higher-order Taylor–Hood discretisations therefore provides confidence in Underworld3 implementations intended for annulus and spherical-shell mantle convection applications.

6 Discussion

The present results demonstrate that Underworld3 reproduces the analytical annulus benchmark solution with convergence behaviour consistent with finite-element theory. The benchmark additionally highlights several important numerical considerations.

First, higher-order mixed discretisations provide substantially improved pressure accuracy relative to lower-order formulations. This is particularly important in geodynamics applications where pressure gradients strongly influence buoyancy-driven flow and dynamic topography.

Second, curved geometries remain sensitive to geometric approximation order. Although linear geometry representation was sufficient to recover theoretical convergence trends for the present benchmark, higher-order curved geometry representations may become increasingly important for extremely fine meshes or strongly curved spherical-shell problems.

Third, equal-order discretisations continue to exhibit weaker pressure robustness relative to stable Taylor–Hood formulations. This behaviour aligns with expectations from the Ladyzhenskaya–Babuska–Brezzi stability condition.

The benchmark therefore provides a valuable verification framework not only for Underworld3, but also for future development of curved-element geometries, advanced boundary-condition implementations, and spherical-shell finite-element formulations.

7 Conclusions

We have presented a verification study of the Thieulot–Puckett annulus Stokes benchmark using Underworld3. The benchmark provides a stringent test of incompressible Stokes flow in curved geometries through the combination of radial gravity, coordinate-dependent forcing, tangential boundary velocity conditions, and non-trivial pressure variation on curved annulus boundaries.

The principal findings are:

1. Higher-order Taylor–Hood discretisations reproduce the expected smooth-solution convergence hierarchy.

2. The $P_2 \times P_1$ discretisation achieves approximately third-order velocity and second-order pressure convergence.

3. The $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity and third-order pressure convergence before fine-resolution error saturation.

4. Lower-order discretisations exhibit weaker pressure robustness and reduced convergence quality.

5. Underworld3 accurately reproduces analytical Stokes solutions in curved annular geometries.

These results establish the Thieulot–Puckett annulus benchmark as an effective verification framework for Underworld3 annulus and spherical-shell geodynamics applications.

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